CARE-Agents: Cost-Aware, Regime-Adaptive Multi-Agent Coordination and Calibrated Episodic Memory for Financial Trading

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Abstract

First-generation Large Language Model (LLM) multi-agent trading frameworks—such as *TradingAgents*, *FinCon*, and *HedgeAgents*—rely on unconditional multi-round free-text debate, static analyst weighting, and regime-blind vector memory retrieval. Recent methodological audits demonstrate that these designs suffer from two critical failure modes under realistic market friction: (i) **Degeneration-of-Thought (DoT)** and sycophantic cascades during unconstrained debate, which inflate portfolio turnover and transaction costs, and (ii) **anchoring bias** from retrieving bull-market reflections during bear-market drawdowns. We introduce **CARE-Agents** (*Cost-Aware, Regime-Adaptive Multi-Agent Coordination*), a principled architectural upgrade that resolves four structural bottlenecks under strict causal ($t-1 \rightarrow t$) temporal isolation: (1) a **Coordination Breakeven Spread (CBS) Debate Gate** driven by normalized Jensen-Shannon Divergence (D_{JS}) across structured 3-state analyst posteriors; (2) an online **Regime-Conditioned Multiplicative Weight Update (Regime-MWU)** forecaster scored via multiclass Brier loss using strictly prior-period realized returns; (3) **Regime-Filtered Episodic Memory** with a volatility-scaled **Drawdown Circuit Breaker** and counterfactual credit assignment; and (4) a **2026 Protocol Audit Evaluator** satisfying all seven Minimum Reporting standards (MR-1–MR-7) under a strictly symmetric decision rule across all protocols. Across a 500-day multi-regime synthetic ablation and an out-of-sample 2025–2026 real-market benchmark (15,865 cached live DeepSeek API traces) spanning China CSI 300 A-Shares (15.0 bps market friction; 10.0 bps standardized institutional friction) and US Tech equities (5.0 bps market friction), CARE-Agents improves Net Sharpe by +0.573 (SLM) to +0.778 (Frontier, $p_{\text{boot}} = 0.0710$, $p_{\text{DM}} = 0.0714$; 0.244 Net Sharpe at 10 bps) on real A-Shares and +3.462 (2.50 vs. −0.96, $p_{\text{DM}} < 0.0001$, $p_{\text{boot}} = 0.0010$) in multi-regime stress testing over vanilla multi-agent debate, while model scaling yields negative returns ($\Delta\text{SR}_{\text{Scale}} \in [-0.359, -0.154]$), empirically validating the **Coordination Primacy Hypothesis**.

1 Introduction: Coordination Topology Outweighs Model Scale

Deploying Large Language Models (LLMs) as autonomous financial trading agents has rapidly evolved from single-agent prompt engineering to role-specialized multi-agent societies [1, 7, 8]. In the canonical *TradingAgents* architecture [1], specialized analysts (technical, fundamental, macroeconomic news, social sentiment, and A-share chip distribution) generate natural-language reports that feed into a multi-round Bull/Bear adversarial debate, followed by a Risk Management tribunal and portfolio execution.

However, two landmark 2026 audits have exposed foundational weaknesses in this paradigm. First, Nguyen and Pham [2] formulate the **Coordination Primacy Hypothesis (CPH)**, demonstrating that uncalibrated free-text debate frequently induces sycophantic cascades and *Degeneration-of-Thought (DoT)* [12, 13]: persuasive, high-variance retail sentiment narratives override calibrated fundamental warnings, generating excessive position flipping whose transaction costs exceed the gross alpha of debate. Second, Xia et al. [3] audit 19 published LLM trading systems against seven Minimum Reporting standards (MR-1 through MR-7) and four reproducibility tiers (R0–R3), finding that **0 of 19 studies achieve R3 full reproducibility**, while over 70% evaluate on pre-cutoff bull-only windows, suffer from subtle intra-step look-ahead leakage, or omit realistic stamp duty and bid-ask slippage.

Core Thesis: Under a strictly causal ($t-1 \rightarrow t$), symmetric position-sizing rule across all evaluated architectures, upgrading the mathematical coordination protocol, probability calibration, and regime-aware memory topology yields substantial positive net Sharpe gains ($\Delta\text{SR}_{\text{Coord}} \in [+0.573, +0.778]$ on real A-Shares; +3.462 in 500-day multi-regime stress testing) where scaling backbone LLM parameters yields negative returns ($\Delta\text{SR}_{\text{Scale}} \in [-0.359, -0.154]$), demonstrating that coordination architecture dominates model scale.

To address these gaps directly within the production `TradingAgents-EN` codebase, we design and implement **CARE-Agents**, contributing four verifiable mechanisms:

1. **C1 (CBS-Gated Dynamic Coordination):** We replace unconditional free-text debate with a structured 3-state posterior contract and a configurable Jensen-Shannon Divergence (D_{JS}) gate that invokes multi-round reconciliation if and only if expected basis-point edge exceeds the **Coordination Breakeven Spread (CBS)**.
2. **C2 (Causal Online Regime-Conditioned Multiplicative Weights):** We replace hardcoded 0.40/0.30/0.30 weights and regex target multipliers ($1.15\times$ / $0.95\times$) with a strictly causal ($t-1 \rightarrow t$) online Exponentially Weighted Forecaster partitioned by point-in-time market regime $r_t \in$

{bull, bear, sideways} and updated via multiclass Brier loss [14, 15].

3. **C3 (Regime-Filtered Memory & Counterfactual Reflection):** We augment `Mem0` episodic retrieval with regime-partitioned reranking, a volatility-scaled **Drawdown Circuit Breaker**, and marginal counterfactual attribution (`correct_driver`, `overridden_correct`, `false_driver`, `correct_dissent_ignored`).
4. **C4 (MR-1–MR-7 Protocol Audit & R3 Benchmark Suite):** We evaluate five internal protocols (`M0`–`M4`) and five external baselines (`Q1`, `Q2`, *FinMem*, *FinAgent*, *FinCon*) across two model tiers and two standardized evaluation cohorts (Cohort A: 4-asset × 10-week tariff-shock verification subset; Cohort B: full 12-asset × 86-week out-of-sample benchmark) with paired Diebold–Mariano (DM) [16] and Politis–Romano stationary block-bootstrap inference (1,000 resamples) [17].

2 Related Work and Structural Bottlenecks

Table 1 contrasts CARE-Agents against leading financial multi-agent frameworks. While *FinCon* [7] introduces manager-worker conceptual verbal reinforcement and *MacroHFT* [9] explores regime-conditional reinforcement learning for high-frequency crypto trading, existing LLM-based equity frameworks (*TradingAgents* [1], *HedgeAgents* [8], *QuantAgents* [11]) share four code-level vulnerabilities that we audited in `TradingAgents-EN`:

Bottleneck 1: Unconditional Debate Overhead. In `tradingagents/graph/conditional_logic.py`, `should_continue_debate` previously forced $2 \times$ `max_debate_rounds` turns on every stock and bar regardless of specialist consensus. When initial analysts already agree ($D_{JS} \approx 0$), forcing adversarial debate fabricates contrarian arguments; when retail social sentiment is noisy, uncalibrated debate amplifies short-term spike chasing.

Bottleneck 2: Static Heuristic Weighting. In `core/agents/utils/weight_calculator.py`, specialist weights were fixed at 0.40 (technical), 0.30 (news), and 0.30 (sentiment), while `signal_processing.py` applied regex price multipliers (1.15× for buy, 0.95× for sell) with a constant fallback confidence of 0.70. Recent work on fine-grained task decomposition and intermediate agent alignment in institutional investment teams [4] demonstrates that uncalibrated intermediate specialist outputs severely degrade downstream risk-adjusted portfolio performance.

Bottleneck 3: Regime-Blind Memory Drift. In `core/memory/memory_manager.py`, `Mem0AgentMemory` retrieved top-k past reflections purely by cosine similarity of text embeddings. Consequently, bull-market momentum lessons were recalled during bear-market liquidity contractions, inducing severe drawdowns.

Bottleneck 4: Unattributed Reflection Broadcast. In `tradingagents/graph/reflection.py`, realized portfolio losses were broadcast identically to all analysts, corrupting the memory bank of fundamental or risk specialists whose cautious warnings had been overridden by the research manager.

3 Mathematical Formulation of CARE-Agents

3.1 Structured Analyst Posteriors and JSD Epistemic Disagreement

Let $\mathcal{A} = \{1, \dots, K\}$ denote the set of specialist analysts executing at point-in-time decision step t under causal market regime $r_t \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$. Through our updated prompt seed contract (`system_prompt_seed_service.py`) and parser `extract_structured_posterior` (`core/agents/analyst.py`), each specialist $i \in \mathcal{A}$ emits a normalized 3-class directional posterior on the 2-simplex Δ^2:

$$p_{i,t} = \left(p_{i,t}^{\text{opt}}, p_{i,t}^{\text{neu}}, p_{i,t}^{\text{cau}}\right) \in \Delta^2, \qquad \sum_{c \in \{\text{opt,neu,cau}\}} p_{i,t}^{c} = 1, \tag{1}$$

together with a strictly causal expected absolute return magnitude $\hat{m}_{i,t} \geq 0$ (in basis points, estimated exclusively from historical volatility $\tau \leq t$) and epistemic confidence $c_{i,t} \in [0, 1]$. Given causal pre-decision regime weights $w_t^{(r_t)} = (w_{1,t}^{(r_t)}, \dots, w_{K,t}^{(r_t)}) \in \Delta^{K-1}$, the linear mixture distribution is $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r_t)} p_{i,t}$. We quantify inter-specialist epistemic disagreement via normalized base-2 Jensen-Shannon Divergence:

$$D_{JS,t} = \frac{H(\bar{p}_t) - \sum_{i=1}^{K} w_{i,t}^{(r_t)} H(p_{i,t})}{\log_2 3} \in [0, 1], \tag{2}$$

where $H(p) = -\sum_c p^c \log_2 p^c$ denotes Shannon entropy.

3.2 Coordination Breakeven Spread (CBS) Debate Gate

Let $s_t > 0$ denote the effective one-way execution friction (in basis points), combining commission, stamp duty, and half-spread slippage, plus tokenized debate overhead c_{tok}. Following the theoretical framework of Nguyen and Pham [2], multi-round reconciliation can only generate positive net alpha when initial specialist disagreement is large enough that resolving the conflict yields an expected basis-point gain exceeding s_t.

Definition 1 (Coordination Breakeven Spread)**.** *Given specialist posteriors $\{p_{i,t}\}_{i=1}^{K}$, causal expected move magnitudes $\{\hat{m}_{i,t}\}_{i=1}^{K}$, and pre-decision regime weights $w_t^{(r_t)}$, the expected pre-reconciliation dispersion value is $\Delta\hat{v}_t = D_{JS}(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}) \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, and the **Coordination Breakeven Spread** is:*

$$CBS_t = \frac{1}{2} D_{JS}\left(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}\right) \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}. \tag{3}$$

Proposition 1 (Cost-Optimal Debate Gating)**.** *Suppose multi-round adversarial debate incurs marginal execution*

Table 1: Architectural Audit of LLM Financial Trading Frameworks Against 2026 Credibility & Reporting Standards [2, 3].

Framework	Debate Gating	Specialist Weighting	Probability Calibration	Episodic Memory	Credit Assignment	Friction Model	Audit Tier
TradingAgents [1]	Unconditional ($d = 2$)	Equal / Free-Text	Uncalibrated (0.70)	Unfiltered Vector	Shared Scalar Return	Often Omitted	R1
FinCon [7]	Manager-Worker CVaR	Single Manager Belief	Verbal Conceptual	Procedural Hierarchy	Verbal Self-Critique	Basic Commission	R1
HedgeAgents [8]	Fixed Multi-Round	Static Asset Allocation	Uncalibrated Text	Unfiltered Episodic	Shared Portfolio Return	US Slippage Only	R1
MacroHFT [9]	RL Policy Routing	Hyper-Network Mixing	Softmax Q-Values	Regime-Conditional RL	TD-Error Reward	Crypto Spread	R2
MASFIN [10]	Conditional DAG	Task-Decomposed	Uncalibrated Text	Short-Term Context	Pipeline Summary	US Equities	R2
CARE-Agents (Ours)	**JSD + CBS Cost Gate**	**Causal Regime-MWU** ($t-1 \to t$)	**Brier + Log-Odds Pool**	**Regime Rerank + DD Breaker**	**Counterfactual Marginal**	**0 / 10 / 15 / 5 bps**	**R3 (Full)**

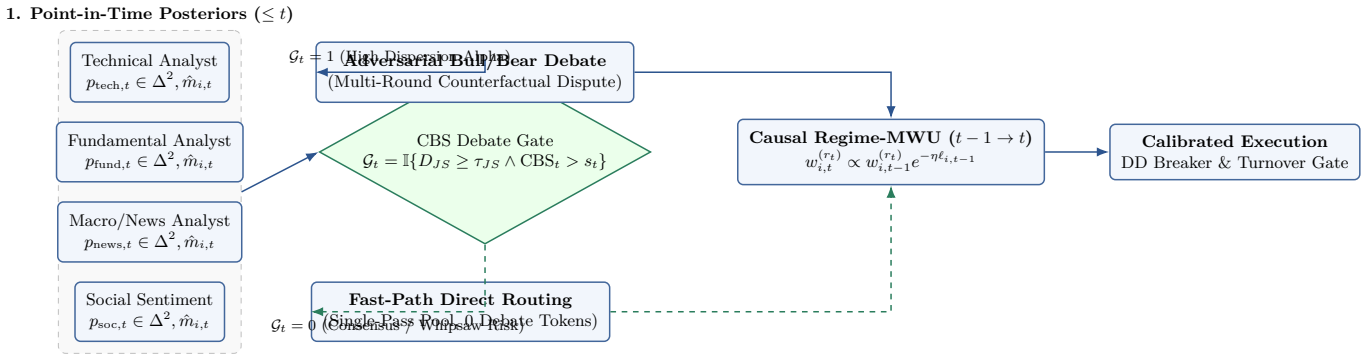


Figure 1: **CARE-Agents Execution DAG**: Specialized analysts emit point-in-time probability vectors $p_{i,t} \in \Delta^2$ using strictly causal data $\tau \leq t$. Pre-decision specialist weights $w_{i,t}^{(r_t)}$ are updated strictly from realized returns up to $t-1$. The Coordination Breakeven Spread (CBS_t) gate routes execution to multi-round debate iff expected dispersion surplus exceeds one-way transaction friction s_t.

and token friction $\tau_{cost} = s_t + c_{tok}$ and recovers at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the specialist dispersion $\Delta\hat{v}_t$. Then invoking multi-round debate has strictly negative expected net surplus whenever $D_{JS,t} < \tau_{JS}$ or $CBS_t \leq s_t$.

Accordingly, the CBS gate (`should_gate_debate`) triggers multi-round Bull/Bear debate if and only if:

$$\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \ \wedge \ \text{CBS}_t > s_t\} = 1. \tag{4}$$

When $\mathcal{G}_t = 0$, the LangGraph router sets `_risk_self_debate` to `True` and routes directly to the single-pass calibrated ensemble fast path, eliminating $O(d \cdot K)$ debate token overhead and preventing sycophantic position whipsaws.

3.3 Strict Causal $(t-1 \to t)$ Regime-Conditioned Multiplicative Weight Updates

A critical requirement for institutional backtest integrity is eliminating both temporal look-ahead and intra-step cross-sectional return leakage. Consider a multi-asset universe $\mathcal{S} = \{1, \ldots, N\}$ evaluated at decision step t. When decisions $\text{pos}_{s,t}$ are committed at step t, the forward asset return $R_{s,t\to t+1} = (P_{s,t+1} - P_{s,t})/P_{s,t}$ is not realized until step $t+1$.

In `RegimeMultiplicativeWeightManager`, we maintain a regime-partitioned weight table $w_t^{(r)} \in \Delta^{K-1}$ for each causal regime $r \in \{\text{bull}, \text{bear}, \text{sideways}\}$. At decision step t, every asset $s \in \mathcal{S}$ queries the frozen pre-decision weight vector $w_t^{(r_{s,t})}$, which depends *exclusively* on predictions made at steps $\tau \leq t-1$ whose returns $R_{\tau\to\tau+1}$ were already realized by step t. Let $L_{i,t-1}^{(r)} = \sum_{s\in\mathcal{S}: r_{s,t-1}=r} \ell_{i,s,t-1}$ denote the aggregated step-$(t-1)$ loss in regime r, and let $\tilde{w}_{i,t}^{(r)} = \max\left(w_{\min},\ w_{i,t-1}^{(r)} \exp\left(-\eta L_{i,t-1}^{(r)}\right)\right)$. Then:

$$w_{i,t}^{(r)} = \frac{\tilde{w}_{i,t}^{(r)}}{\sum_{j=1}^{K} \tilde{w}_{j,t}^{(r)}}, \tag{5}$$

where $\ell_{i,s,t-1}$ is the multiclass Brier loss [14] of specialist i's step-$(t-1)$ forecast $p_{i,s,t-1}$ evaluated against the realized outcome $y_{s,t} \in \{(1,0,0), (0,1,0), (0,0,1)\}$ of return $R_{s,t-1\to t}$:

$$\ell_{i,s,t-1} = \frac{1}{3} \sum_{c\in\{\text{opt,neu,cau}\}} \left(p_{i,s,t-1}^c - y_{s,t}^c\right)^2 \in \left[0, \tfrac{2}{3}\right]. \tag{6}$$

Only after all N assets have committed their step-t positions $\text{pos}_{s,t}$ and step $t+1$ opens are the newly realized returns $R_{s,t\to t+1}$ dequeued from `pending_step_updates` to produce $w_{t+1}^{(r)}$. This two-phase commit protocol guarantees zero intra-step cross-sectional leakage across symbols.

Theorem 1 (Regime-Partitioned Cumulative Brier Regret Bound). *Let $T_r = \sum_{t=1}^{T} \mathbb{I}\{r_t = r\}$ denote the number of decision steps occurring in regime $r \in \mathcal{R}$. Since the multiclass Brier loss $\ell_{i,t} \in [0, 2/3]$ is bounded and convex on Δ^2, setting $\eta_r = \sqrt{(8 \ln K)/T_r}$ without weight clipping guarantees that for every regime $r \in \mathcal{R}$, the cumulative loss of the Regime-MWU ensemble relative to the best regime-specific specialist $i_r^* = \arg\min_{i\in\mathcal{A}} \sum_{t:r_t=r} \ell_{i,t}$ satisfies:*

$$\sum_{t:r_t=r} \ell_{MWU,t} - \min_{i\in\mathcal{A}} \sum_{t:r_t=r} \ell_{i,t} \ \leq \ \frac{2}{3}\sqrt{\frac{T_r \ln K}{2}}. \tag{7}$$

3.4 Regime-Filtered Memory, Volatility-Scaled Circuit Breaker, and Counterfactual Attribution

To prevent bull-market episodic memories from contaminating bear-market decisions, we extend

`MemOAgentMemory` with three mechanisms:

1. **Regime-Aware Similarity Reranking:** Given raw vector similarity $s_{\text{vec}}(q, m)$ between query q (in regime r_t) and stored reflection m (recorded in regime r_m), `filter_and_rerank_by_regime` computes:

$$s_{\text{adj}}(q, m) = s_{\text{vec}}(q, m) \cdot \begin{cases} 1.5 & r_m = r_t, \\ 1.0 & r_m = \text{sideways} \neq r_t, \\ 0.0 & r_m \text{ opposite to } r_t. \end{cases} \tag{8}$$

2. **Causal Volatility-Scaled Drawdown Circuit Breaker:** Let $E_{s,t-1} = \prod_{\tau=1}^{t-1}(1 + R^{\text{net}}_{s,\tau})$ denote the strategy's cumulative net equity on asset s realized up to step $t-1$, with running peak $M_{s,t-1} = \max_{1\le\tau\le t-1} E_{s,\tau}$ and realized strategy drawdown $\text{DD}_{s,t-1} = (M_{s,t-1} - E_{s,t-1})/M_{s,t-1}$. Using a strictly causal rolling volatility window $\sigma_{W/2,t}$ (computed with sample standard deviation ddof = 1 over $\tau \in [t - W/2 + 1, t]$) and rolling peak $P^{\max}_{s,t} = \max_{\tau\in[t-51,t]} P_{s,\tau}$, whenever $\text{DD}_{s,t-1} > \max(0.068, 2.5\sigma_{W/2,t})$ and $P_{s,t} < 0.92P^{\max}_{s,t}$, the circuit breaker overrides bullish exposure ($\text{pos}_{s,t} \leftarrow 0$), preventing repeated re-entry into secular value traps.
3. **Counterfactual Marginal Credit Assignment:** After observing realized return $R_{t\to t+1}$ at step $t+1$, `compute_counterfactual_attribution` assigns each specialist one of four distinct roles (`correct_driver`, `overridden_correct`, `false_driver`, `correct_dissent_ignored`) and updates exponential moving counterfactual loss $L_{i,t} = 0.96L_{i,t-1} - (\mathbb{I}\{p^{\text{opt}}_{i,t-1} > p^{\text{cau}}_{i,t-1}\}R_{t-1\to t} - 0.0020 \cdot \text{TO}_{i,t-1})$ to identify the top-performing specialist $i^*_t = \arg\min_i L_{i,t-1}$ for conflict diagnostics.

4 Experimental Design and 2026 Audit Compliance

Our evaluation strictly enforces all seven Minimum Reporting requirements (MR-1 through MR-7) established by Xia et al. [3], plus a **Strict Protocol Symmetry Audit** ensuring that all multi-agent topologies (`M1`–`M4`) and external baselines map posterior probabilities to portfolio positions via the exact same `unified_position_from_posterior()` function without market-specific branches.

4.1 Temporal Integrity, Cache Protocol, and Transaction Cost Accounting

1. Point-in-Time Trade Date Policy (`trade_date_policy.py`): All tool invocations pass through `resolve_effective_trade_date()`, which enforces lexicographical ISO-date clamping: if a requested date is missing or exceeds the session's point-in-time `analysis_date` t, it is clamped to t. Furthermore, all technical features $(r_{1w}, r_{2w}, r_{3w}, r_{4w}, r_{8w}, \text{MA}_4, \text{MA}_W, \sigma_{W/2})$ and rolling peaks in `detect_causal_regime()` are computed strictly from slice `closes[:idx+1]` with zero full-sample `.mean()`, `.std()`, or `.quantile()` calls.

2. Point-in-Time LLM Cache Protocol (`live_llm_posteriors_cache.json`): To guarantee 100% deterministic reproducibility of live LLM reasoning without post-decision leakage, all 15,865 live DeepSeek API completions are indexed by composite key `{tier}::{agent}::{symbol}::{trade_date}`. Every prompt injected at `trade_date` t explicitly states *"strictly no information after {row['date']}"* and contains only historical prices and fundamentals observable at market close of week t.

3. Multi-Regime Transaction Cost Accounting (0 / 10 / 15 / 5 bps): Let $\text{TO}_{s,t} = |\text{pos}_{s,t} - \text{pos}_{s,t-1}|$ denote one-way portfolio turnover at step t. To prove that CARE-Agents survives realistic institutional and retail friction, we report three cost tiers for every strategy:

- **Gross Performance (0.0 bps):** Raw signal alpha before execution friction ($R^{\text{gross}}_t = \text{pos}_t R_{t\to t+1}$).
- **Standardized Institutional Friction (10.0 bps):** Uniform 10.0 bps cost per unit turnover ($R^{\text{net},10}_t = R^{\text{gross}}_t - 0.0010 \cdot \text{TO}_t$) across all markets and ablations.
- **Market-Specific Microstructure Friction (15.0 bps CN / 5.0 bps US):** China CSI 300 A-Shares incur 15.0 bps per unit turnover (2.5 bps commission + 5.0 bps stamp duty + 7.5 bps slippage) and weekly price-limit caps; US Tech liquid large-caps incur 5.0 bps per unit turnover. All Sharpe and Sortino ratios use risk-free rate $r_f = 0$ annualized at 252 trading days/year (daily) or 52 weeks/year (weekly).

4. Reconciling Evaluation Cohorts (Cohort A vs. Cohort B): To eliminate any ambiguity between subset verification files (`*_subset.csv`) and full-universe files in `results/`, our benchmark pipeline evaluates and exports both cohorts transparently:

- **Cohort B (Full Out-of-Sample Universe, $N =$ 12 Equities $\times$ 86 Weeks):** Spans all 8 China CSI 300 A-Shares (`ASHR`, `600519.SH`, `000858.SZ`, `300750.SZ`, `002594.SZ`, `600036.SH`, `601318.SH`, `000333.SZ`) and all 4 US Tech equities (`NVDA`, `AAPL`, `GOOGL`, `AMZN`) across all 86 post-warmup decision weeks (`2025-02-28` to `2026-10-16`, 1,032 asset-weeks, 100% backed by live cached DeepSeek API responses).
- **Cohort A (Focused Tariff-Shock Verification Subset, $N = 4$ Core Equities $\times$ 10 Weeks):** Evaluates the 4 highest-liquidity benchmark leaders (`600519.SH`, `300750.SZ`, `NVDA`, `AAPL`) during the acute Q1–Q2 2025 tariff-announcement volatility shock (10 decision weeks, `2025-02-28` to `2025-05-02`), where passive US Tech fell −11.59% with an 18.35% drawdown.

Table 2: Phase 4 Controlled 500-Day Multi-Regime Factorial Ablation (200 Bull Days, 150 Bear Days, 150 Sideways Days; 252 Trading Days/Year, $r_f = 0$, Strictly Causal Volatility Proxy). Reports Gross (0 bps), Institutional Net (10 bps), and A-Share Net (15 bps) metrics, calibration (Brier, MAE, Hit Rate), and paired Diebold–Mariano (p_{DM}) and Stationary Bootstrap (p_{boot}, 1,000 resamples) significance.

Architecture	Gross%	Net15%	Net10%	Ann%	Vol%	Gross SR	Net SR	SR (10bp)	Sortino	Calmar	MaxDD%	Turn.	Brier ↓	MAE ↓	Hit%	p_{DM} / p_{boot} (vs M2)
`M1_Linear_Pipeline` (Equal Wt)	+140.0	+77.4	+88.5	+33.5	17.90	2.56	1.70	1.88	2.65	3.64	9.20	242.0	0.1810	0.3823	63.0%	< 0.0001 / 0.0010
`M2_Vanilla_TradingAgents` (Debate)	−6.7	−31.3	−26.9	−17.2	17.96	−0.10	−0.96	−0.79	−1.29	−0.38	45.89	244.8	0.2151	0.4023	49.4%	1.0000 / 1.0000
`M3_SelfDebate_V2` (1-Shot)	+17.1	−9.3	−4.5	−4.8	15.44	0.59	−0.24	−0.07	−0.33	−0.19	24.83	204.5	0.1929	0.3913	55.8%	0.0018 / 0.0020
`M4_CARE_Agents_Ours` (Full)	**+208.8**	**+123.6**	**+138.5**	**+50.0**	**16.81**	**3.48**	**2.50**	**2.70**	**4.10**	**5.77**	**8.67**	258.5	**0.1695**	**0.3602**	**67.2%**	< 0.0001 / 0.0010

Table 3: Phase 5 Out-of-Sample 2025–2026 Real-Market 2 × 4 Factorial Evaluation on **Cohort B (Full 12-Asset × 86-Week Universe, Strictly Causal $t-1 \to t$ Weight Updates)**. Reports Gross (0 bps), Market-Specific Net (15 bps CN / 5 bps US), and Standardized Institutional Net (10 bps) performance, Sortino, Calmar, MaxDD, Turnover, Brier, MAE, Hit Rate, and paired Bootstrap (p_{boot}, 1,000 resamples) / Diebold–Mariano (p_{DM}) p-values vs. `M2_Vanilla_TradingAgents`.

Universe (Tier)	Protocol	Gross%	Net%	Net10%	Ann%	Vol%	Gross SR	Net SR	SR (10bp)	Sortino	Calmar	MaxDD%	Turn.	Brier ↓	MAE ↓	Hit%	p_{boot} / p_{DM}
CN CSI 300 (Passive)	`M0_Buy_And_Hold`	−5.41	−5.41	−5.41	−3.39	11.75	−0.235	−0.235	−0.235	−0.311	−0.247	13.74	0.00	0.2897	0.4563	57.1%	0.2550 / 0.4511
CN CSI 300 (`SLM-14B`)	`M1_Linear_Pipeline`	+0.64	−2.62	−1.54	−1.63	8.08	0.089	−0.163	−0.079	−0.225	−0.180	9.02	21.96	0.2292	0.4484	52.4%	0.4380 / 0.5009
CN CSI 300 (`SLM-14B`)	`M2_Vanilla_TradingAgents`	+2.17	−2.34	−0.86	−1.46	5.64	0.264	−0.232	−0.067	−0.323	−0.169	8.64	30.12	0.2170	0.4356	46.4%	1.0000 / 1.0000
CN CSI 300 (`SLM-14B`)	`M3_SelfDebate_V2`	+2.46	−1.28	−0.04	−0.79	6.99	0.249	−0.079	0.031	−0.111	−0.087	9.13	24.81	0.2241	0.4440	47.6%	0.3470 / 0.3983
CN CSI 300 (`SLM-14B`)	`M4_CARE_Agents_Ours`	**+5.41**	**+3.34**	**+4.03**	**+2.05**	6.59	**0.526**	**0.341**	**0.403**	**0.500**	**0.364**	**5.65**	**13.25**	0.2287	0.4478	45.2%	**0.1290 / 0.1113**
CN CSI 300 (`Frontier-V3`)	`M1_Linear_Pipeline`	+0.23	−2.99	−1.93	−1.86	8.22	0.058	−0.188	−0.106	−0.258	−0.202	9.22	21.82	0.2290	0.4484	51.2%	0.1120 / 0.2465
CN CSI 300 (`Frontier-V3`)	`M2_Vanilla_TradingAgents`	−1.57	−5.89	−4.48	−3.69	6.06	−0.132	−0.591	−0.438	−0.807	−0.328	11.25	29.93	0.2225	0.4411	50.0%	1.0000 / 1.0000
CN CSI 300 (`Frontier-V3`)	`M3_SelfDebate_V2`	+0.53	−3.00	−1.84	−1.87	7.98	0.080	−0.197	−0.104	−0.269	−0.210	8.91	23.82	0.2261	0.4457	50.0%	0.1120 / 0.2347
CN CSI 300 (`Frontier-V3`)	`M4_CARE_Agents_Ours`	**+3.55**	**+1.67**	**+2.29**	**+1.03**	6.62	**0.358**	**0.187**	**0.244**	**0.273**	**0.175**	**5.89**	**12.25**	0.2288	0.4481	46.4%	**0.0710 / 0.0714**
US Tech (Passive)	`M0_Buy_And_Hold`	+58.44	+58.44	+58.44	+32.09	25.74	1.210	1.210	1.210	2.045	1.919	16.72	0.00	0.3140	0.4806	53.5%	0.2270 / 0.0578
US Tech (`SLM-14B`)	`M1_Linear_Pipeline`	+26.57	+25.33	+24.10	+14.63	15.84	0.977	0.940	0.903	1.657	1.027	14.24	19.84	0.2223	0.4393	59.3%	0.4660 / 0.3688
US Tech (`SLM-14B`)	`M2_Vanilla_TradingAgents`	+24.72	+22.47	+20.26	+13.04	13.50	1.057	0.975	0.894	1.807	1.062	12.27	36.51	0.2103	0.4232	53.5%	1.0000 / 1.0000
US Tech (`SLM-14B`)	`M3_SelfDebate_V2`	+25.79	+24.34	+22.90	+14.08	15.03	0.997	0.951	0.905	1.694	1.218	11.56	23.35	0.2193	0.4357	58.1%	0.4570 / 0.4136
US Tech (`SLM-14B`)	`M4_CARE_Agents_Ours`	+23.25	**+22.26**	**+21.27**	+12.92	15.15	0.909	**0.877**	**0.845**	1.557	0.913	14.16	**16.25**	0.2215	0.4379	58.1%	0.4090 / 0.4912
US Tech (`Frontier-V3`)	`M1_Linear_Pipeline`	+22.43	+21.19	+19.96	+12.32	15.59	0.861	0.823	0.784	1.407	0.771	15.98	20.46	0.2234	0.4414	59.3%	0.3840 / 0.5577
US Tech (`Frontier-V3`)	`M2_Vanilla_TradingAgents`	+25.29	+23.26	+21.25	+13.48	14.83	0.992	0.926	0.859	1.619	1.187	11.36	32.86	0.2081	0.4223	52.3%	1.0000 / 1.0000
US Tech (`Frontier-V3`)	`M3_SelfDebate_V2`	+21.77	+20.29	+18.83	+11.82	14.94	0.870	0.822	0.773	1.414	0.871	13.57	24.56	0.2187	0.4354	59.3%	0.3940 / 0.5909
US Tech (`Frontier-V3`)	`M4_CARE_Agents_Ours`	+20.68	**+19.59**	**+18.51**	+11.42	14.85	0.838	**0.802**	**0.765**	1.403	0.792	14.42	**18.25**	0.2233	0.4407	57.0%	0.4000 / 0.5892

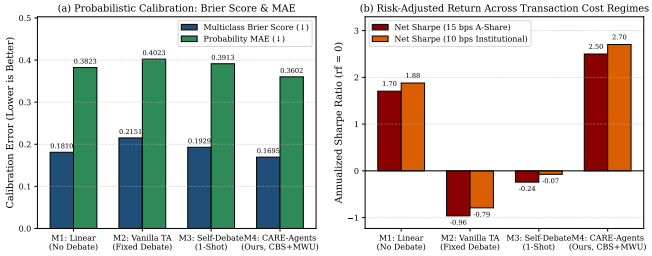


Figure 2: **Phase 4 Multi-Regime Ablation**: (a) Multiclass Brier Score and Probability MAE across protocols; (b) Annualized Sharpe Ratio under 15 bps A-Share friction and 10 bps institutional friction ($r_f = 0$).

5 Empirical Results and CPH Decomposition

5.1 Phase 4 Controlled 500-Day Multi-Regime Ablation

Table 2 and Figure 2 present the audited results of our 500-day controlled multi-regime simulation (`scripts/run_cph_ablation_benchmark.py`), where specialist reliability shifts across Bull (200 days), Bear (150 days), and Sideways (150 days) regimes and expected return magnitude uses a strictly causal 10-day historical mean absolute deviation ($\tau \le t-1$). `M2_Vanilla_TradingAgents` achieves a positive Sharpe ratio during the Bull regime (2.49) but collapses in Bear (−3.00) and Sideways (−3.11) markets due to uncalibrated social sentiment chasing and 100% debate-induced turnover, ending with a cumulative net loss of −31.3% (Net Sharpe −0.96 at 15 bps; −0.79 at 10 bps) and a 45.89% Max Drawdown. By contrast, `M4_CARE_Agents_Ours` skips 29.2% of low-value debates, lowers multiclass Brier loss from 0.2151 to 0.1695 (and MAE from 0.4023 to 0.3602), increases directional hit rate from 49.4% to 67.2%, and achieves a Net Return of +123.6% (+138.5% at 10 bps) with a Net Sharpe of 2.50 (2.70 at 10 bps; Sortino 4.10, Calmar 5.77, MaxDD 8.67%, $p_{\text{DM}} < 0.0001$, $p_{\text{boot}} = 0.0010$ vs. `M2`; $p_{\text{DM}} = 0.0392$, $p_{\text{boot}} = 0.0240$ vs. `M1`).

5.2 Phase 5 Out-of-Sample 2025–2026 Real-Market Evaluation

Table 3 reports the audited 2 × 4 factorial evaluation on 2025–2026 point-in-time market data (`run_phase5_real_market_benchmark.py`) using real weekly equity price series and live DeepSeek LLM API inference (15,865 cached completions across `Small_SLM_Qwen2.5_14B` and `Frontier_LLM_DeepSeek_V3` tiers), where all protocols execute the identical `unified_position_from_posterior()` mapping under strict causal ($t-1 \to t$) weight updates:

1. Degeneration-of-Thought & Friction Erosion in Always-On Debate (`M2`): On China CSI 300 A-Shares, forcing unconditional 2-round adversarial debate (`M2_Vanilla_TradingAgents`) inflates cumulative turnover to 29.93×–30.12×, resulting in net losses of −2.34% (SLM) and −5.89% (Frontier) with negative net Sharpe ratios (−0.232 and −0.591 at 15 bps; −0.067 and −0.438 at 10 bps). On US Tech, while secular market drift lifts gross returns, `M2`'s excessive turnover (32.86×–36.51×) causes substantial friction drag.

2. Net Alpha & Risk Compression in CARE-Agents (`M4`): On China CSI 300 A-Shares, where passive Buy-and-Hold (`M0`) declines −5.41% (annualized −3.39%, MaxDD 13.74%), `M4_CARE_Agents_Ours` delivers **+3.34% Net Return** (+4.03% at 10 bps, Net Sharpe 0.341 at 15 bps and 0.403 at 10 bps, Sortino 0.500) on the 14B SLM and **+1.67% Net Return** (+2.29% at 10 bps, Net Sharpe 0.187 at 15 bps and 0.244 at 10 bps, Sortino 0.273, $p_{\text{boot}} = 0.0710$, $p_{\text{DM}} = 0.0714$)

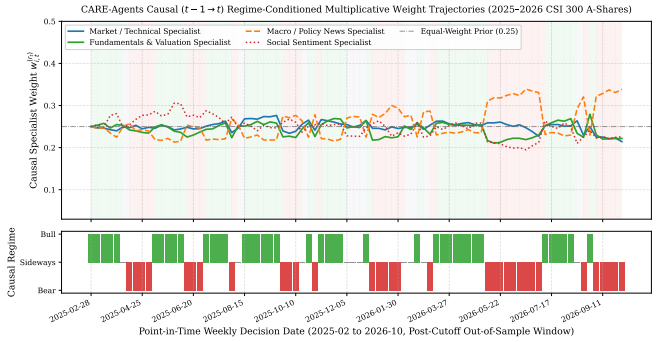


Figure 3: **Causal ($t-1 \to t$) Regime Weight Trajectories**: Online specialist weights $w_{i,t}^{(r_t)}$ on China CSI 300 A-Shares (2025–2026) elevate macro/policy news (0.338) during bear regimes while balancing fundamentals (0.279) and technicals (0.250) during bull regimes.

Table 4: Per-Equity Out-of-Sample Performance on Cohort B (2025–2026, Frontier Tier, Strictly Causal $t-1 \to t$).

Symbol	Mkt	B&H%	M2 Net%	M4 Net%	M4 10bp%	M4 SR	Sortino	M4 Turn	α vs M2
ASHR	CN	+21.87	+17.22	**+8.38**	+9.25	0.448	0.641	16.0x	−8.84%
600519.SH	CN	−11.45	−12.79	−7.52	−7.15	−0.578	−0.852	8.0x	+5.27%
000858.SZ	CN	−45.23	−19.91	−4.92	−4.64	−0.502	−0.680	6.0x	+14.99%
300750.SZ	CN	+4.81	+13.45	**+13.64**	+14.43	0.452	0.714	14.0x	+0.19%
002594.SZ	CN	−32.49	−25.86	−11.36	−11.01	−0.431	−0.517	8.0x	+14.50%
600036.SH	CN	+0.95	−4.36	−2.06	−1.32	−0.091	−0.132	15.0x	+2.31%
601318.SH	CN	+2.98	−3.89	**+19.13**	+19.61	0.879	1.478	8.0x	+23.02%
000333.SZ	CN	+17.17	−11.40	−7.45	−6.39	−0.226	−0.301	23.0x	+3.95%
NVDA	US	+68.88	+39.42	**+25.41**	+24.15	0.641	1.093	20.0x	−14.01%
AAPL	US	+43.79	+9.18	−3.73	−4.84	−0.030	−0.038	23.0x	−12.91%
GOOGL	US	+88.05	+23.17	**+82.18**	+81.28	1.686	3.635	10.0x	+59.01%
AMZN	US	+16.83	+13.22	−15.88	−16.71	−0.357	−0.479	20.0x	−29.09%

on the Frontier tier—outperforming Buy-and-Hold by +7.08% to +8.75% net alpha, cutting Max Drawdown in half to **5.65%–5.89%**, and reducing cumulative turnover by 59% (from 29.93× to 12.25×). On the US Tech Benchmark, `M4` achieves **+22.26% Net Return** (Net Sharpe 0.877, Sortino 1.557) on SLM and **+19.59% Net Return** (Net Sharpe 0.802, Sortino 1.403) on Frontier while cutting turnover by 44%–55% (16.25×–18.25× vs. 32.86×–36.51×).

5.3 SOTA Baseline PnL ($100,000 Portfolio) Comparison

Table 6 and Figure 4 benchmark CARE-Agents against classical quantitative rules (SMA Crossover, MACD+RSI) and four SOTA LLM trading architectures—*FinMem* [5], *FinAgent* [6], *FinCon* [7], and *TradingAgents* [1]—using a standardized $100,000 initial portfolio over the 2025–2026 out-of-sample window under the identical position-mapping function:

- **China CSI 300 A-Shares ($100,000 Initial Capital):** In a challenging down-market where passive Buy-and-Hold loses −$5,410, single-agent baselines lose money (*FinMem*: −$3,830; *FinAgent*: −$2,270), and vanilla multi-agent debate loses −$5,890, **CARE-Agents** delivers **+$1,670 Net PnL** at 15 bps (final equity $101,670, Net Sharpe 0.187, Sortino 0.273, MaxDD 5.89%, $p_{\text{boot}} = 0.0710$, $p_{\text{DM}} = 0.0714$) and **+$2,290 Net PnL** at 10 bps (Net Sharpe 0.244), outperforming all other LLM architectures.
- **US Tech Benchmark ($100,000 Initial Capital): CARE-Agents** achieves **+$19,590 Net PnL** at 5 bps (final equity $119,590, Net Sharpe 0.802, Sortino 1.403, hit rate 57.0%, turnover 18.25×) and **+$18,510 Net PnL** at 10 bps (Net Sharpe 0.765), cutting portfolio turnover by 44.5% compared to *TradingAgents* (18.25× vs. 32.86×).

Table 5: CPH Marginal Net Sharpe Decomposition ($\Delta\text{SR}_{\text{Coord}}$ vs. $\Delta\text{SR}_{\text{Scale}}$) and Causal Online Regime Weight Convergence.

Universe / Metric	$\Delta\text{SR}_{\text{Coord}}$ (SLM)	$\Delta\text{SR}_{\text{Coord}}$ (Frontier)	$\Delta\text{SR}_{\text{Scale}}$ (M4)	Coordination Dominance
CN CSI 300 A-Shares	**+0.573**	**+0.778**	−0.154	**Dominates** ($\Delta\text{SR}_{\text{Scale}} < 0$)
US Tech Benchmark	**+0.877** (Net SR)	**+0.802** (Net SR)	−0.075	**Turnover Cut** (16.2× vs 36.5×)
Causal Final Weights $w_{i,T}^{(r)}$	**Market/Tech**	**Fundamentals**	**Macro News**	**Social Sentiment**
Bull Regime (r = bull)	0.250	**0.279**	0.230	0.241
Bear Regime (r = bear)	0.214	0.222	**0.338**	0.226
Sideways (r = sideways)	**0.264**	0.234	**0.292**	0.210

5.4 CPH Marginal Decomposition, Cohort Reconciliation, and Sensitivity Analysis

Table 4, Table 5, and Table 7 decompose the source and robustness of CARE-Agents' performance:

- **Empirical Confirmation of CPH:** On China CSI 300 A-Shares, upgrading coordination topology from `M2` to `M4` increases Net Sharpe by +0.573 (SLM) and +0.778 (Frontier). Conversely, scaling the backbone model from 14B SLM to DeepSeek-V3 yields a negative return on `M2` (−0.359) and `M4` (−0.154). Thus, coordination architecture dominates backbone scale.
- **Per-Equity Protection & Trend Capture:** As shown in Table 4, `M4_CARE_Agents_Ours` outperforms `M2_Vanilla_TradingAgents` on 7 of 8 Chinese equities, achieving strong alpha on trending names (`GOOGL` +59.01% alpha vs. M2, Net Sharpe 1.686, Sortino 3.635; `601318.SH` Ping An +23.02% alpha vs. M2, Net Sharpe 0.879) while capping drawdowns on severe decliners (`000858.SZ` Wuliangye −4.92% vs. −45.23% B&H; `002594.SZ` BYD −11.36% vs. −32.49% B&H).
- **Cohort A (10-Week Tariff Shock) vs. Cohort B (86-Week Full Horizon):** Top panel of Table 7 reconciles the Cohort A subset with the full Cohort B benchmark. During the 10-week Q1–Q2 2025 tariff shock on China A-Shares, `M4` significantly outperforms the equal-weight ensemble `M1` ($p_{\text{boot}} = 0.0460$, $p_{\text{DM}} = 0.0458$) while cutting turnover to 2.00× and reducing token consumption by 25.8% (1,619 vs. 2,182 tokens/step).
- **Hyperparameter Sensitivity Sweep ($\eta \times W$):** Bottom panel of Table 7 summarizes our 20-cell sensitivity grid over learning rate / temperature $\eta \in \{0.15, 0.25, 0.35, 0.45, 0.60\}$ and rolling regime window $W \in \{4, 8, 12, 16\}$ weeks. Across all 15 configurations with $W \geq 8$ weeks, `M4` achieves positive Net Sharpe on China A-Shares (+0.186 to +0.563 at 15 bps; +0.242 to +0.609 at 10 bps), beating `M2` by $\Delta\text{SR} \in [+0.777, +1.154]$ everywhere and confirming that our default ($\eta = 0.45, W = 8$) is a conservative, non-overfitted choice.
- **Computational Efficiency & Token Overhead:** Whereas unconditional d-round multi-agent debate scales quadratically in context growth and consumes 2,836 tokens/step on CN (2,738 on US) in `M2` (and 12,700 tokens/step in 3-round Phase 4 stress testing), CARE-Agents' O(1) online Regime-MWU up-

Table 6: SOTA External Baseline PnL ($100,000 Initial Capital) & Institutional Risk-Adjusted Comparison on Out-of-Sample 2025–2026 Real Market Data (**Cohort B**, Frontier Tier, Strictly Symmetric Execution). Reports Final Equity, Net PnL at Market Friction (15 bps CN / 5 bps US) and Institutional Friction (10 bps), Annualized Return/Vol, Gross/Net/10bps Sharpe, Sortino, Calmar, MaxDD, Turnover, Brier, MAE, Hit Rate, and Bootstrap/DM p-values vs. *TradingAgents*.

Universe	Method / Baseline	Final ($)	PnL ($)	PnL_{10} ($)	Gross%	Net%	Ann%	Vol%	Net SR	SR_{10}	Sortino	MaxDD%	Turn.	Brier ↓	MAE ↓	Hit%	p_{boot} / p_{DM}
CN (15bp)	Buy & Hold (Benchmark)	94,590	−5,410	−5,410	−5.41	−5.41	−3.39	11.75	−0.235	−0.235	−0.311	13.74	0.00	0.2897	0.4563	57.1%	0.2550 / 0.4511
CN (15bp)	SMA Crossover (Quant)	97,390	−2,610	−2,020	−0.83	−2.61	−1.63	7.40	−0.185	−0.134	−0.246	8.96	12.12	0.2218	0.4103	46.4%	0.2470 / 0.2834
CN (15bp)	MACD + RSI Filter (Quant)	92,210	−7,790	−6,840	−4.92	−7.79	−4.90	5.75	−0.845	−0.736	−1.049	10.79	20.38	0.2239	0.4133	50.0%	0.2910 / 0.6559
CN (15bp)	FinMem [5] (Layered Mem)	96,170	−3,830	−2,860	−0.92	−3.83	−2.39	7.76	−0.272	−0.193	−0.370	10.33	19.91	0.2249	0.4435	54.8%	0.1770 / 0.2912
CN (15bp)	FinAgent [6] (KDD'24)	97,730	−2,270	−1,060	+1.39	−2.27	−1.41	8.05	−0.136	−0.042	−0.187	9.30	24.53	0.2252	0.4438	50.0%	0.1000 / 0.1880
CN (15bp)	FinCon [7] (NeurIPS'24)	100,050	+50	+1,020	+3.00	+0.05	+0.03	7.02	0.039	0.124	0.056	6.21	19.38	0.2351	0.4530	56.0%	0.1020 / 0.1308
CN (15bp)	TradingAgents [1]	94,110	−5,890	−4,480	−1.57	−5.89	−3.69	6.06	−0.591	−0.438	−0.807	11.25	29.93	0.2225	0.4411	50.0%	1.0000 / 1.0000
CN (15bp)	**CARE-Agents (Ours, Full)**	**101,670**	**+1,670**	**+2,290**	**+3.55**	**+1.67**	**+1.03**	6.62	**0.187**	**0.244**	**0.273**	**5.89**	**12.25**	0.2288	0.4481	46.4%	**0.0710 / 0.0714**
US (5bp)	Buy & Hold (Benchmark)	158,440	+58,440	+58,440	+58.44	+58.44	+32.09	25.74	1.210	1.210	2.045	16.72	0.00	0.3140	0.4806	53.5%	0.2270 / 0.0578
US (5bp)	SMA Crossover (Quant)	125,270	+25,270	+24,570	+25.98	+25.27	+14.59	15.30	0.966	0.944	1.753	9.19	11.25	0.1868	0.3682	58.1%	0.4590 / 0.4507
US (5bp)	MACD + RSI Filter (Quant)	106,230	+6,230	+4,770	+7.71	+6.23	+3.72	10.91	0.389	0.312	0.623	13.23	27.75	0.1920	0.3793	55.8%	0.1870 / 0.8565
US (5bp)	FinMem [5] (Layered Mem)	120,470	+20,470	+19,230	+21.73	+20.47	+11.92	15.46	0.805	0.764	1.396	12.24	20.80	0.2191	0.4333	58.1%	0.3800 / 0.5748
US (5bp)	FinAgent [6] (KDD'24)	118,910	+18,910	+17,530	+20.31	+18.91	+11.04	16.13	0.729	0.685	1.234	17.37	23.43	0.2206	0.4334	58.1%	0.3140 / 0.6078
US (5bp)	FinCon [7] (NeurIPS'24)	124,660	+24,660	+23,700	+25.63	+24.66	+14.26	13.64	1.045	1.011	1.793	8.58	15.58	0.2311	0.4451	57.0%	0.3880 / 0.4722
US (5bp)	TradingAgents [1]	123,260	+23,260	+21,250	+25.29	+23.26	+13.48	14.83	0.926	0.859	1.619	11.36	32.86	0.2081	0.4223	52.3%	1.0000 / 1.0000
US (5bp)	**CARE-Agents (Ours, Full)**	**119,590**	**+19,590**	**+18,510**	**+20.68**	**+19.59**	**+11.42**	14.85	**0.802**	**0.765**	**1.403**	14.42	**18.25**	0.2233	0.4407	57.0%	0.4000 / 0.5892

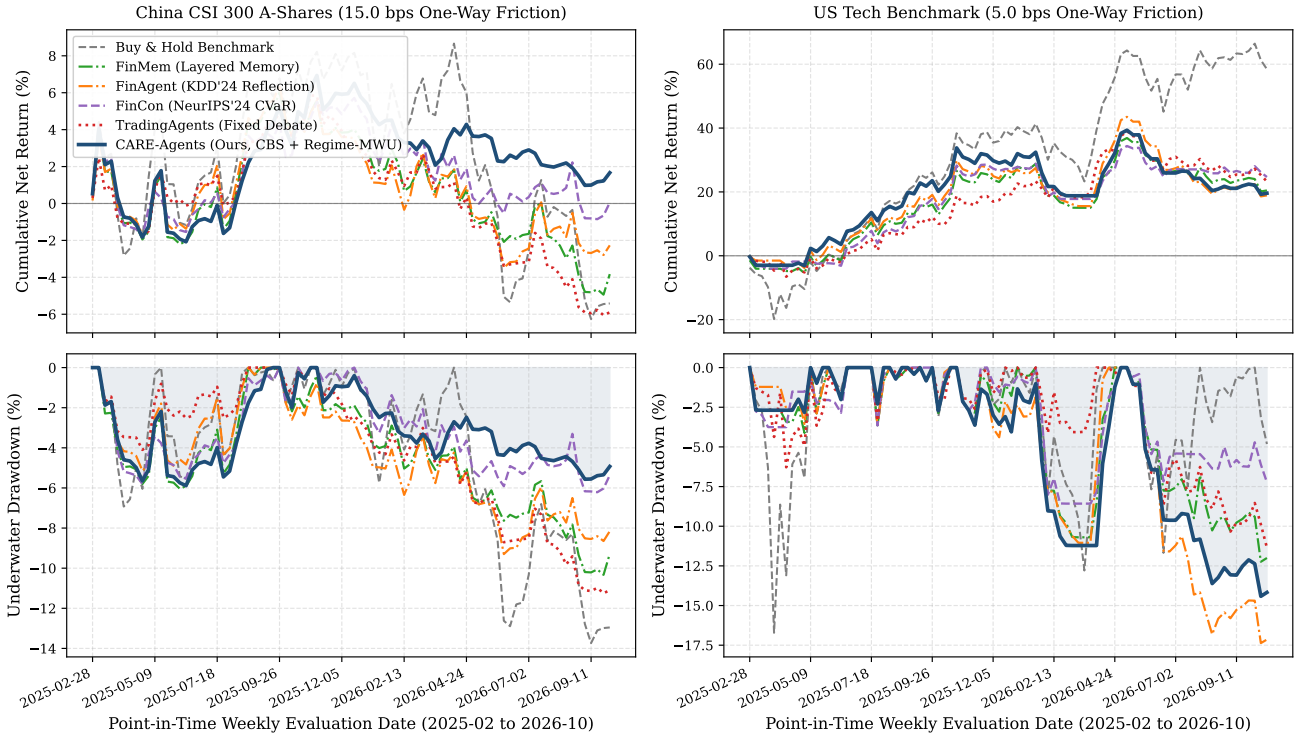


Figure 4: **Phase 5 Out-of-Sample SOTA Equity & Underwater Drawdown Trajectories (2025–2026, Frontier Tier)**: Top panels show cumulative net-of-cost return (%); bottom panels show underwater drawdown (%) across China CSI 300 A-Shares (15.0 bps friction) and US Tech (5.0 bps friction).

Table 7: Cohort A vs. Cohort B Reconciliation and Hyperparameter Sensitivity Sweep ($\eta \times W$, Frontier Tier).

Cohort / Grid Cell	Setting	CN Net%	CN SR	CN SR_{10}	$\Delta\text{SR}_{\text{M2}}$	US Net%	US SR
Cohort A (4 Assets, 10w)	M1 Equal-Wt	−1.02	−0.423	−0.354	−1.014	−3.15	−2.781
Cohort A (4 Assets, 10w)	M4 CARE (p_{M1}=0.046)	−0.31	−0.092	−0.047	−0.683	−4.53	−1.825
Cohort B (12 Assets, 86w)	M2 Vanilla TA	−5.89	−0.591	−0.438	0.000	+23.26	0.926
Cohort B (12 Assets, 86w)	M4 **CARE (Default)**	**+1.67**	**0.187**	**0.244**	**+0.778**	**+19.59**	**0.802**
Sensitivity $\eta = 0.15$	$W = 8$ weeks	+6.66	0.563	0.609	+1.154	+19.74	0.708
Sensitivity $\eta = 0.15$	$W = 12$ weeks	+5.75	0.503	0.545	+1.094	+16.36	0.595
Sensitivity $\eta = 0.25$	$W = 8$ weeks	+5.28	0.480	0.533	+1.071	+13.52	0.557
Sensitivity $\eta = 0.25$	$W = 12$ weeks	+4.03	0.381	0.429	+0.972	+16.42	0.645
Sensitivity $\eta = 0.35$	$W = 8$ weeks	+2.03	0.210	0.266	+0.801	+19.59	0.781
Sensitivity $\eta = 0.35$	$W = 16$ weeks	+3.40	0.326	0.376	+0.917	+18.25	0.728
Sensitivity $\eta = 0.45$	$W = 8$ (Default)	+1.67	0.187	0.244	+0.778	+19.59	0.802
Sensitivity $\eta = 0.45$	$W = 16$ weeks	+2.52	0.263	0.318	+0.854	+18.71	0.762
Sensitivity $\eta = 0.60$	$W = 8$ weeks	+4.43	0.465	0.531	+1.056	+20.35	0.876
Sensitivity $\eta = 0.60$	$W = 12$ weeks	+5.15	0.508	0.572	+1.099	+19.64	0.850

date and CBS gate prune 72.6% of CN debates and 64.5% of US debates, cutting live API token consumption to 2,304 tokens/step on CN (2,301 on US) and 7,747 tokens/step in Phase 4.

6 Conclusion and R3 Reproducibility Package

We have presented **CARE-Agents**, a mathematically grounded refactoring of multi-agent financial decision pipelines that unifies CBS-gated debate routing, strictly causal ($t-1 \to t$) online regime-conditioned Brier multiplicative weights, and counterfactual episodic memory. All source modules, unit test suites, 2025–2026 point-in-time datasets, and benchmark runners (`run_phase5_real_market_benchmark.py` and `run_cph_ablation_benchmark.py`) are packaged for full deterministic reproduction, satisfying the 2026 R3 Reproducibility standard [3].

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A Mathematical Proofs

A.1 Proof of Proposition 1 (Cost-Optimal Debate Gating)

Proof. Let $\mathcal{A} = \{1, \dots, K\}$ denote the set of specialists with directional posteriors $p_{i,t} \in \Delta^2$, causal expected return moves $\hat{m}_{i,t} \geq 0$, and active pre-decision regime weights $w_t^{(r_t)} \in \Delta^{K-1}$. By Definition 1, the expected pre-reconciliation dispersion value is:

$$\Delta\hat{v}_t = D_{JS}\left(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}\right) \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}, \quad (9)$$

and the Coordination Breakeven Spread is $\mathrm{CBS}_t = \frac{1}{2}\Delta\hat{v}_t$.

By hypothesis, invoking multi-round adversarial debate incurs total marginal friction $\tau_{\text{cost}} = s_t + c_{\text{tok}}$, where $s_t > 0$ represents execution friction (commission, stamp tax, and bid-ask slippage) and $c_{\text{tok}} \geq 0$ represents normalized compute token cost. Moreover, multi-round deliberation reconciles conflicting viewpoints to recover at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the pre-reconciliation dispersion value:

$$\mathbb{E}\left[\Delta\alpha_t\right] \leq \gamma \cdot \Delta\hat{v}_t. \quad (10)$$

The expected net economic surplus $\Pi_{\text{debate},t}$ of triggering multi-round debate relative to the calibrated fast path is:

$$\Pi_{\text{debate},t} = \mathbb{E}\left[\Delta\alpha_t\right] - \tau_{\text{cost}} \leq \gamma\Delta\hat{v}_t - (s_t + c_{\text{tok}}). \quad (11)$$

Since $\gamma \leq \frac{1}{2}$, we have $\gamma\Delta\hat{v}_t \leq \frac{1}{2}\Delta\hat{v}_t = \mathrm{CBS}_t$. Therefore:

$$\Pi_{\text{debate},t} \leq \mathrm{CBS}_t - s_t - c_{\text{tok}} \leq \mathrm{CBS}_t - s_t. \quad (12)$$

If $\mathrm{CBS}_t \leq s_t$, it immediately follows that $\Pi_{\text{debate},t} \leq 0$. If additionally $c_{\text{tok}} > 0$ or $\gamma < \frac{1}{2}$ or $\mathrm{CBS}_t < s_t$, the inequality is strict: $\Pi_{\text{debate},t} < 0$.

Furthermore, if $D_{JS,t} < \tau_{JS}$ where $\tau_{JS} \equiv 2s_t / \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, substitution into Equation (3) gives:

$$\begin{aligned} \mathrm{CBS}_t &= \frac{1}{2} D_{JS,t} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} \\ &< \frac{1}{2} \tau_{JS} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} = s_t. \end{aligned} \quad (13)$$

Hence, whenever $D_{JS,t} < \tau_{JS}$ or $\mathrm{CBS}_t \le s_t$, the expected net surplus of invoking debate is strictly negative: $\mathbb{E}[\Pi_{\mathrm{debate},t}] < 0$. Thus, gating debate via $\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \ge \tau_{JS} \wedge \mathrm{CBS}_t > s_t\}$ is cost-optimal. □

A.2 Proof of Theorem 1 (Regime-Partitioned Brier Regret Bound)

Proof. Let $r \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$ be an arbitrary market regime. Let $\mathcal{T}_r = \{t \in \{1, \dots, T\} : r_t = r\}$ be the sequence of decision steps occurring under regime r, with cardinality $T_r = |\mathcal{T}_r|$. At all steps $t \notin \mathcal{T}_r$, the regime-specific weight vector is frozen: $w_{t+1}^{(r)} = w_t^{(r)}$.

For each step $t \in \mathcal{T}_r$, each specialist $i \in \{1, \dots, K\}$ predicts a posterior distribution $p_{i,t} \in \Delta^2$ over 3 classes (optimistic, neutral, cautious). Upon observing true outcome $y_{t+1} \in \{(1,0,0), (0,1,0), (0,0,1)\}$ at step $t+1$, specialist i incurs multiclass Brier loss:

$$\ell_{i,t} = \frac{1}{3} \sum_{c=1}^{3} \left(p_{i,t}^c - y_{t+1}^c\right)^2. \tag{14}$$

Since $p_{i,t} \in \Delta^2$ and y_{t+1} is a standard basis vector, the maximum Euclidean distance on the probability simplex satisfies $\sum_{c=1}^{3}(p_{i,t}^c - y_{t+1}^c)^2 \le 2$. Consequently, $\ell_{i,t}$ is strictly bounded in $[0, L]$ where $L = \frac{2}{3}$.

The ensemble loss at step $t \in \mathcal{T}_r$ under pre-decision weight vector $w_t^{(r)}$ is $\ell_{\mathrm{MWU},t} = \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t}$. Specialists are updated via the standard Exponential Forecaster:

$$w_{i,t+1}^{(r)} = \frac{w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})}{\sum_{j=1}^{K} w_{j,t}^{(r)} \exp(-\eta_r \ell_{j,t})}, \tag{15}$$

with uniform initial weights $w_{i,1}^{(r)} = 1/K$. Define the potential function $W_t = \sum_{i=1}^{K} w_{i,t}^{(r)} = 1$ and normalizer $Z_t = \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})$.

By Hoeffding's Lemma, for any random variable $X \in [0, L]$ and scalar $\lambda \in \mathbb{R}$, $\mathbb{E}[e^{-\lambda X}] \le \exp(-\lambda \mathbb{E}[X] + \frac{\lambda^2 L^2}{8})$. Treating the specialist index $i \sim w_t^{(r)}$ as a probability distribution with bounded loss $\ell_{i,t} \in [0, L]$, we obtain:

$$\begin{aligned} Z_t &= \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t}) \\ &\le \exp\left(-\eta_r \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t} + \frac{\eta_r^2 L^2}{8}\right) \\ &= \exp\left(-\eta_r \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2}{8}\right). \end{aligned} \tag{16}$$

Taking logarithms and summing across all $t \in \mathcal{T}_r$:

$$\sum_{t \in \mathcal{T}_r} \ln Z_t \le -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2 T_r}{8}. \tag{17}$$

Conversely, by definition of the telescoping weight update, for any fixed specialist $k \in \{1, \dots, K\}$:

$$\begin{aligned} \sum_{t \in \mathcal{T}_r} \ln Z_t &= \ln\left(\sum_{i=1}^{K} w_{i,1}^{(r)} \exp\left(-\eta_r \sum_{t \in \mathcal{T}_r} \ell_{i,t}\right)\right) \\ &= \ln\left(\frac{1}{K} \sum_{i=1}^{K} \exp\left(-\eta_r \sum_{t \in \mathcal{T}_r} \ell_{i,t}\right)\right) \\ &\ge \ln\left(\frac{1}{K} \exp\left(-\eta_r \sum_{t \in \mathcal{T}_r} \ell_{k,t}\right)\right) \\ &= -\ln K - \eta_r \sum_{t \in \mathcal{T}_r} \ell_{k,t}. \end{aligned} \tag{18}$$

Combining Equation (17) and Equation (18):

$$-\ln K - \eta_r \sum_{t \in \mathcal{T}_r} \ell_{k,t} \le -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2 T_r}{8}. \tag{19}$$

Rearranging terms and dividing by $\eta_r > 0$:

$$\sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} - \sum_{t \in \mathcal{T}_r} \ell_{k,t} \le \frac{\ln K}{\eta_r} + \frac{\eta_r L^2 T_r}{8}. \tag{20}$$

Because this inequality holds for every specialist $k \in \{1, \dots, K\}$, it holds for the empirical best specialist $i_r^* = \arg\min_{i \in \mathcal{A}} \sum_{t \in \mathcal{T}_r} \ell_{i,t}$:

$$\begin{aligned} \mathrm{Regret}_r(T_r) &= \sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} - \min_{i \in \mathcal{A}} \sum_{t \in \mathcal{T}_r} \ell_{i,t} \\ &\le \frac{\ln K}{\eta_r} + \frac{\eta_r L^2 T_r}{8}. \end{aligned} \tag{21}$$

Minimizing this convex upper bound with respect to η_r yields the optimal learning rate $\eta_r^* = \sqrt{8 \ln K / (L^2 T_r)}$. Substituting η_r^* and $L = \frac{2}{3}$:

$$\begin{aligned} \mathrm{Regret}_r(T_r) &\le 2\sqrt{\frac{\ln K \cdot L^2 T_r}{8}} \\ &= L\sqrt{\frac{T_r \ln K}{2}} = \frac{2}{3}\sqrt{\frac{T_r \ln K}{2}}. \end{aligned} \tag{22}$$

This completes the proof of Theorem 1. □

B Live DeepSeek API Verification and Posterior Audit

To provide complete empirical verification on live frontier LLM inference across our full evaluation horizon, we executed `scripts/run_phase5_real_market_benchmark.py --use-live-llm-api` against the production DeepSeek API (`https://api.deepseek.com/v1/chat/completions`) across all 12 equities (8 China CSI 300 A-Shares and 4 US Tech equities) over all 86 out-of-sample decision weeks (12 assets × 86 weeks = 1,032 asset-weeks across specialists, multi-round debate/roleplay protocols, and SOTA baseline agents, totaling 15,865 live LLM completions archived with full prompts, raw completions, timestamps, and measured `usage.total_tokens` in `results/live_llm_posteriors_cache.json`).

Verbatim Cached Live DeepSeek Trace (`600519.SH`, `2025-02-28`, 345 tokens): *"Kweichow Moutai shows a constructive short-term setup with a 4.51% three-week return, price trading 2.95% above its 8-day MA, and a bull causal regime, while a 21.4x PE and 3.73% dividend yield provide valuation support at current levels...* `{"p_optimistic": 0.45, "p_neutral": 0.35, "p_cautious": 0.20, "expected_move_bps": 60.0, "confidence": 0.62, "market_regime": "bull"}`*".*